

MOIZ GHAZANFAR ALI

Backend Developer | Database Architecture Enthusiast | AI & ML Explorer

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PROFESSIONAL SUMMARY

Highly motivated Intermediate (Computer Science) student with hands-on experience in backend development, database architecture, and AI-powered systems. Proficient in building REST APIs, designing scalable database schemas, and implementing Retrieval-Augmented Generation (RAG) pipelines. Currently contributing to a real-world product while pursuing formal education, demonstrating strong initiative, quick learning adaptability, and an analytical mindset.

TECHNICAL SKILLS

Languages:	Python, SQL
Frameworks:	FastAPI, LangChain, LangGraph
Databases:	PostgreSQL, MongoDB (Schema Design, Indexing, Query Optimization)
AI / ML:	RAG Pipelines, Machine Learning, Deep Learning, Computer Vision
APIs & Services:	REST API Design, Third-party API Integration
Core Concepts:	Database Architecture, Data Modeling, System Design (Basics)

TECHNICAL PROJECTS

Live Data Analytics Chatbot In Progress

Tech Stack: Python, FastAPI, LangGraph, PostgreSQL, MongoDB

- Developing an intelligent chatbot that connects directly to a live production database to answer real-time business queries (e.g., sales figures, inventory metrics) using natural language.
- Designing a secure architecture to safely translate natural language queries into read-only SQL and MongoDB aggregations in real time.
- Focusing on critical backend requirements including secure database exposure, query injection safety, and a low-latency execution pipeline.

Retrieval-Augmented Generation (RAG) Chatbot Completed

Tech Stack: Python, FastAPI, LangChain, MongoDB

- Engineered a fully functional RAG chatbot that parses, chunks, and indexes documents to answer complex user queries with context-aware precision.
- Designed and optimized the data ingestion and vector storage pipeline utilizing advanced embedding models integrated with MongoDB for metadata management.
- Built and deployed clean, high-performance API endpoints via FastAPI for seamless document uploading, indexing status tracking, and query processing.

School Management System

POC - Completed

Tech Stack: Python, FastAPI, MongoDB

- Designed a comprehensive relational-style database schema covering multi-entity relationships including students, teachers, classes, attendance records, and grades with proper normalization.
- Developed robust RESTful APIs using FastAPI to handle end-to-end CRUD operations integrated with flexible role-based access control (RBAC).
- Engineered the database layer with indexed foreign keys and optimized join queries to ensure quick data retrieval and high system performance.
- **Project Showcase:** [View LinkedIn Post](#)

EDUCATION

Intermediate - Computer Science (ICS)

2025 - 2026

Ziauddin Board, Karachi

(Exams in Progress)